

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1008+050

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

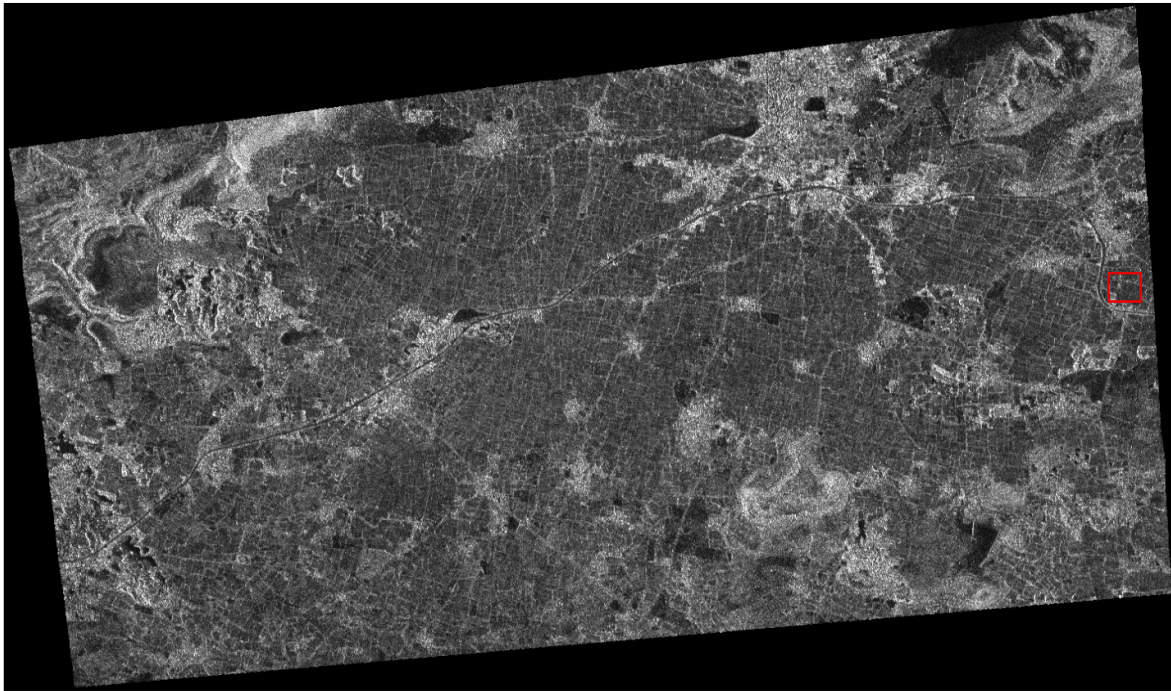
Flight Direction - DESCENDING

Polarization – VV+VH

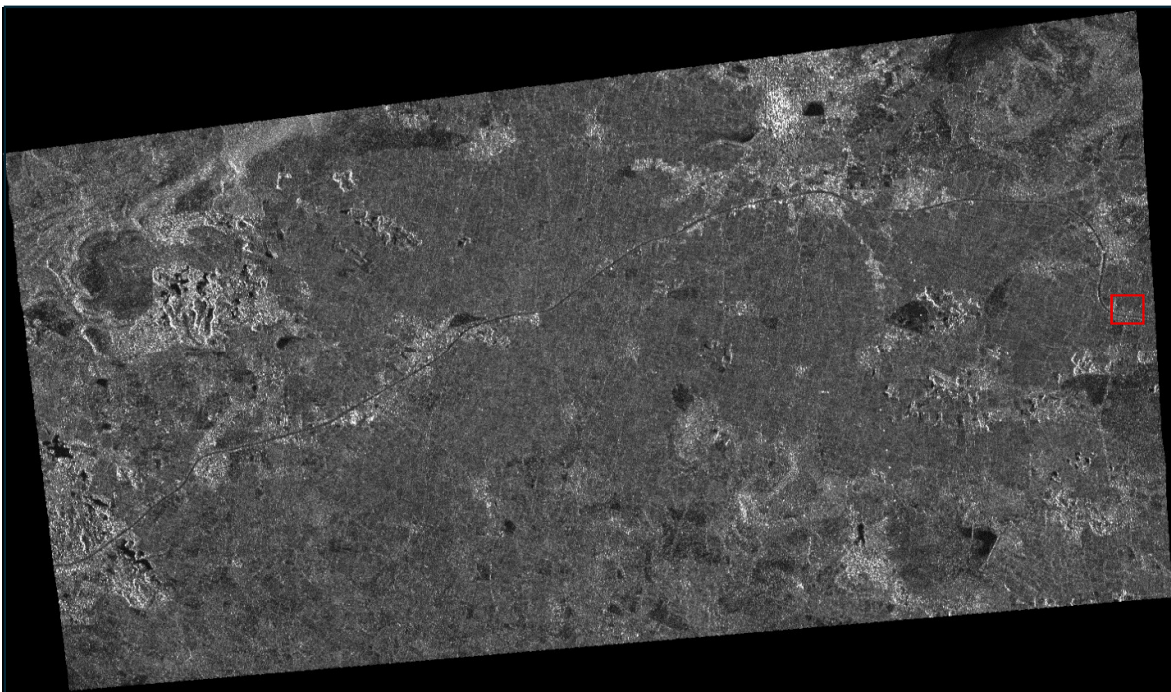
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

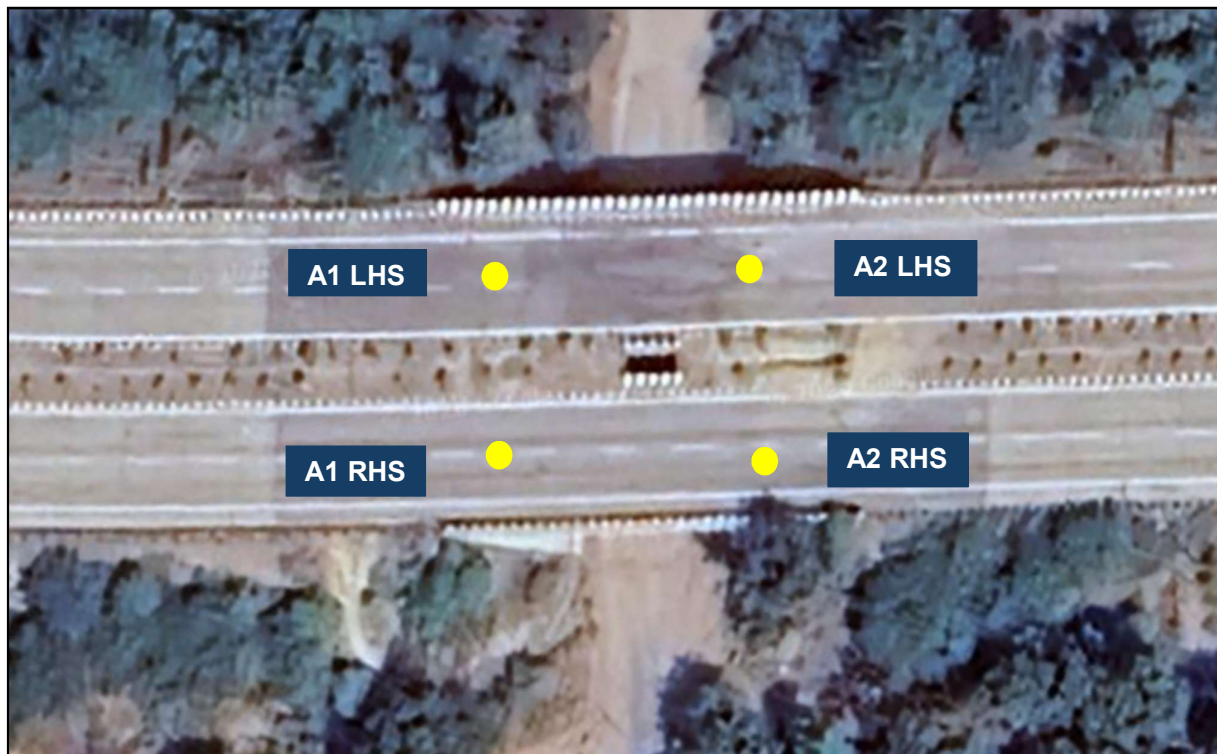


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

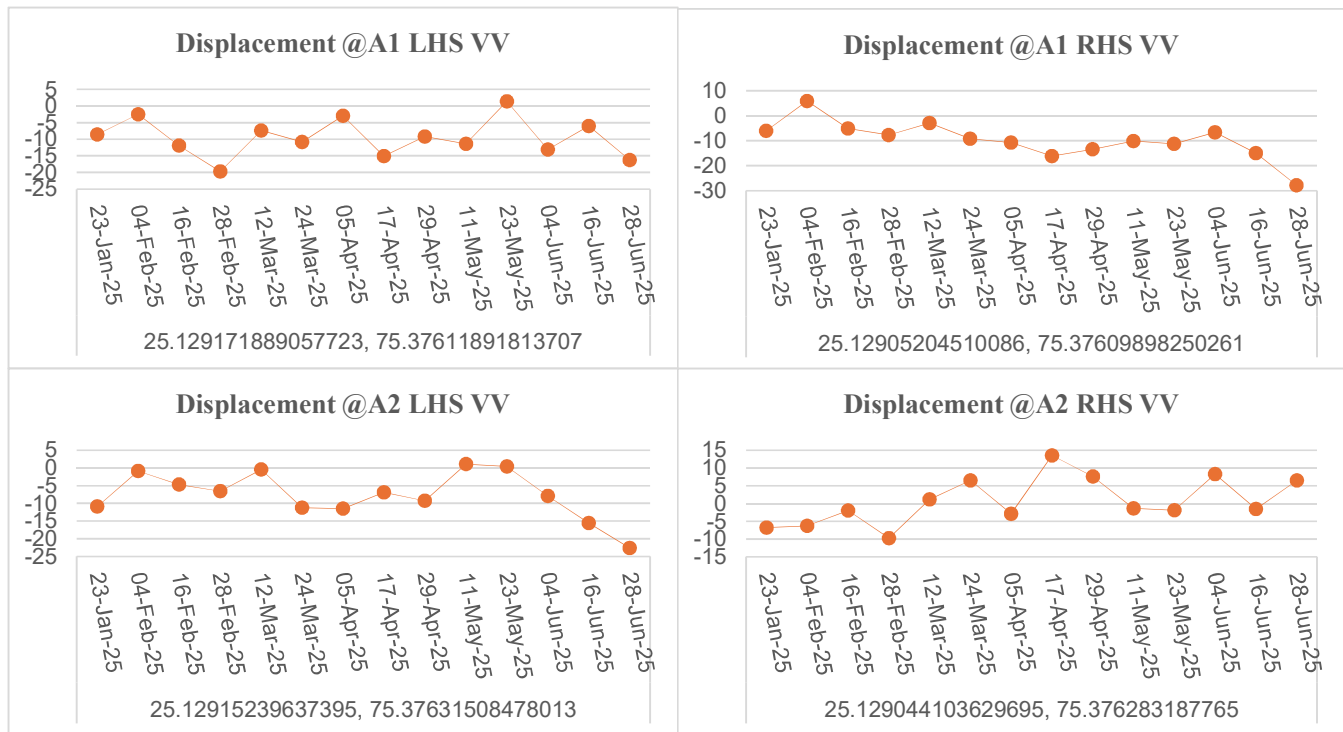
VUP – CH-1008+050

Length – 33.97M

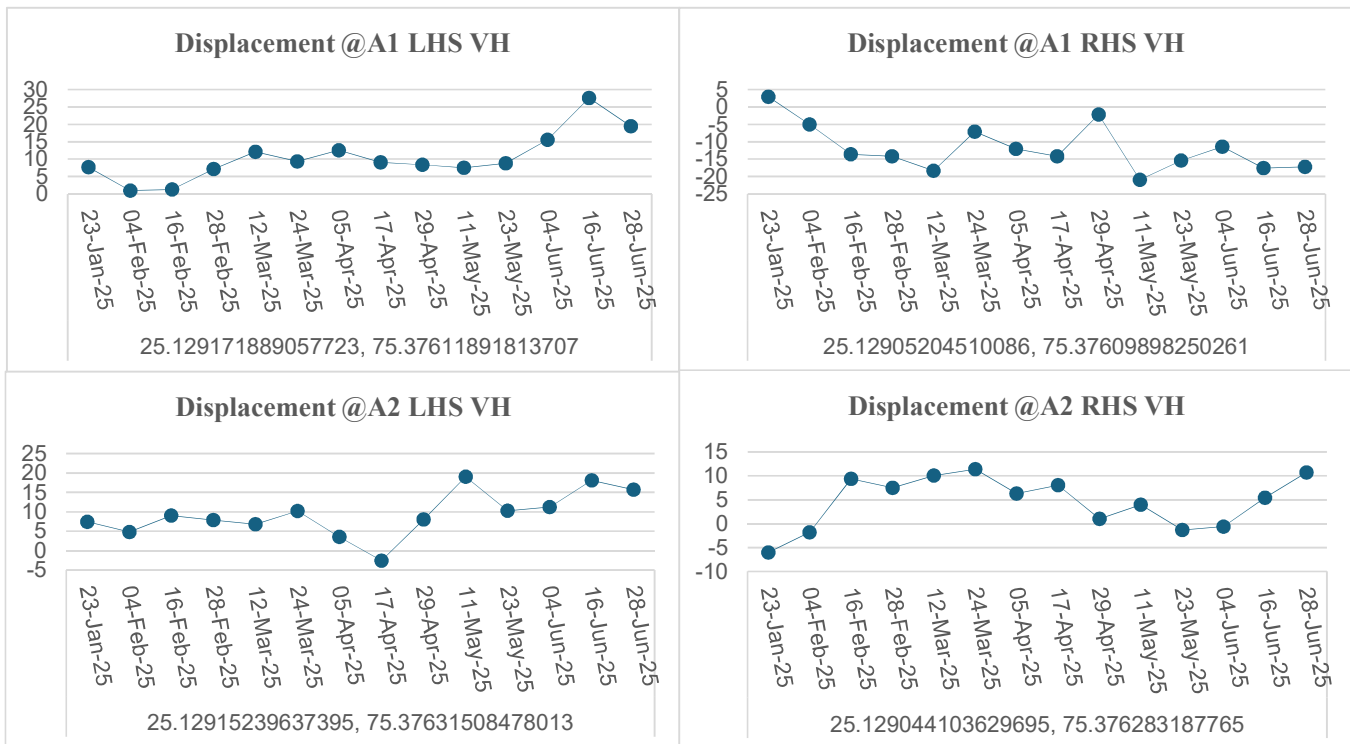


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



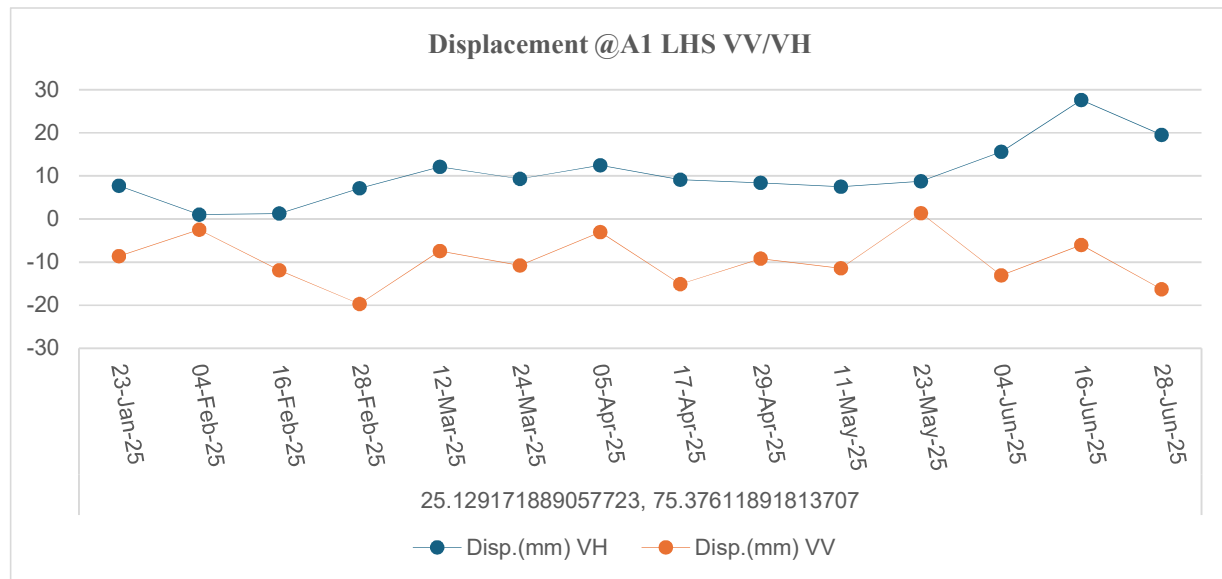
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.129171889057723, 75.37611891813707)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits dominant settlement behaviour, with values ranging roughly between -3 mm and -20 mm.
- **Horizontal Displacement (VH)** at A1 LHS shows a consistent increasing trend, ranging approximately from 0 mm to +27 mm.

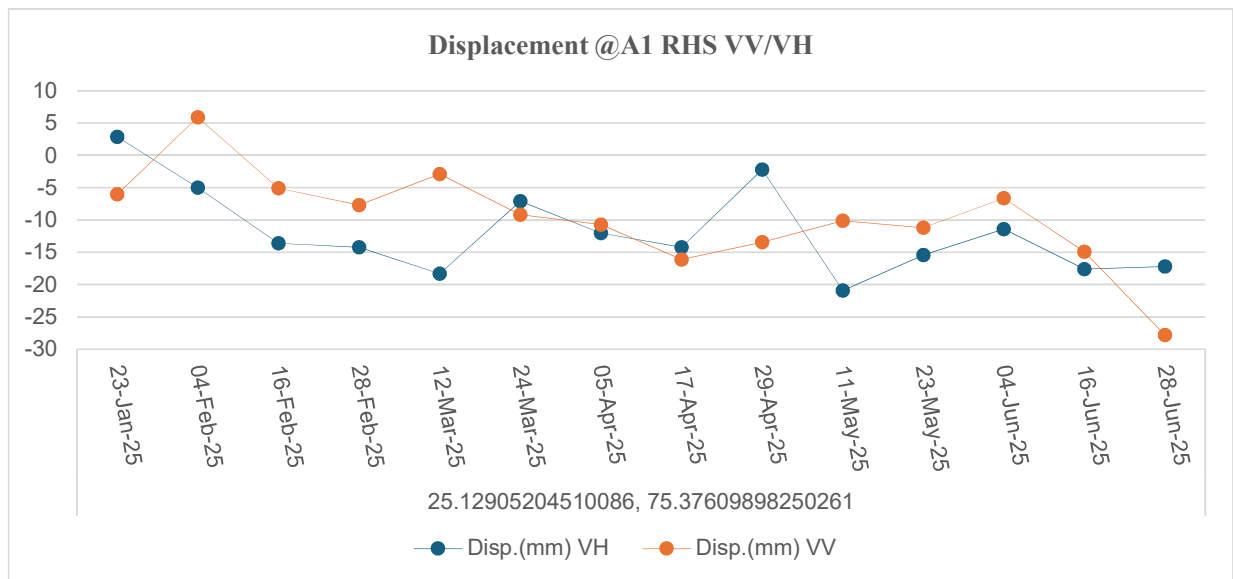
Critical Events & Anomalies:

- **January 2025:** VH begins around +7 mm, while VV is near -9 mm, suggesting initial lateral displacement accompanied by vertical settlement.
- **June 2025:** VH peaks at approximately +27 mm, while VV simultaneously drops back to around -14 mm, highlighting strong lateral displacement dominance with renewed settlement.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.12905204510086, 75.37609898250261)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows high variability with an overall negative trend, ranging approximately from +3 mm to -22 mm.
- **Vertical displacement (VV)** exhibits predominant settlement behaviour, with values ranging roughly between +5 mm uplift and -28 mm settlement.

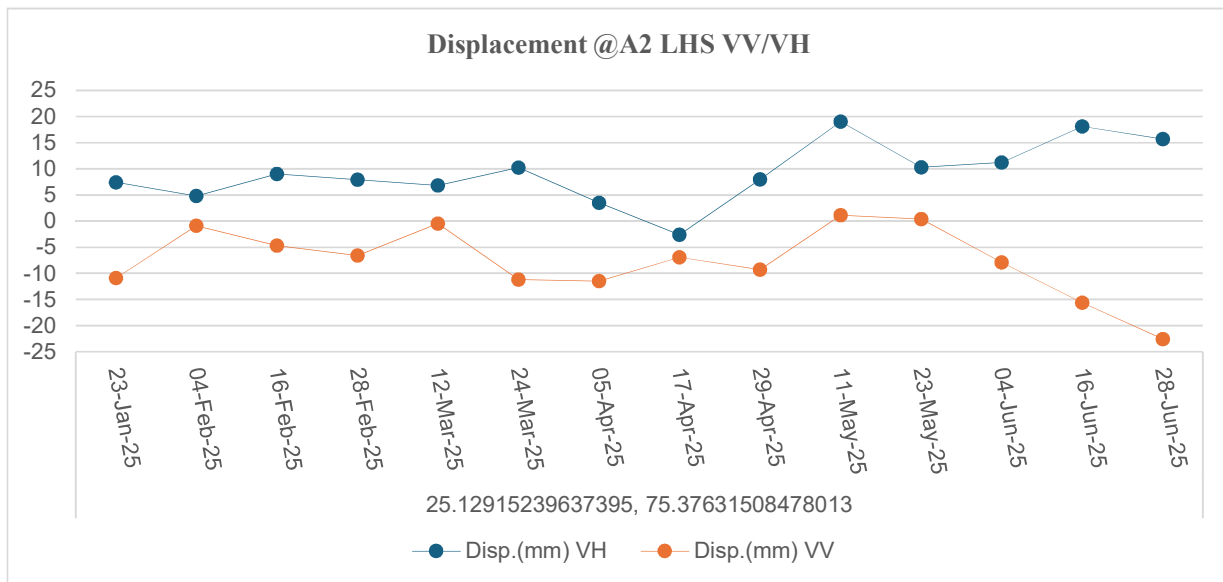
Critical Events & Anomalies:

- **January 2025:** VH starts near +3 mm, while VV is around -6 mm, indicating initial mild lateral displacement accompanied by vertical settlement.
- **June 2025:** VV experiences a sharp settlement event, reaching approximately -28 mm, while VH remains between -12 mm and -17 mm, highlighting severe vertical settlement with sustained lateral displacement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.12915239637395, 75.37631508478013)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS shows moderate to high variability with an overall increasing positive trend, ranging approximately from –2 mm to +19 mm.
- **Vertical Displacement (VV)** exhibits predominantly settlement behaviour, with values ranging roughly from +1 mm uplift to –22 mm settlement.

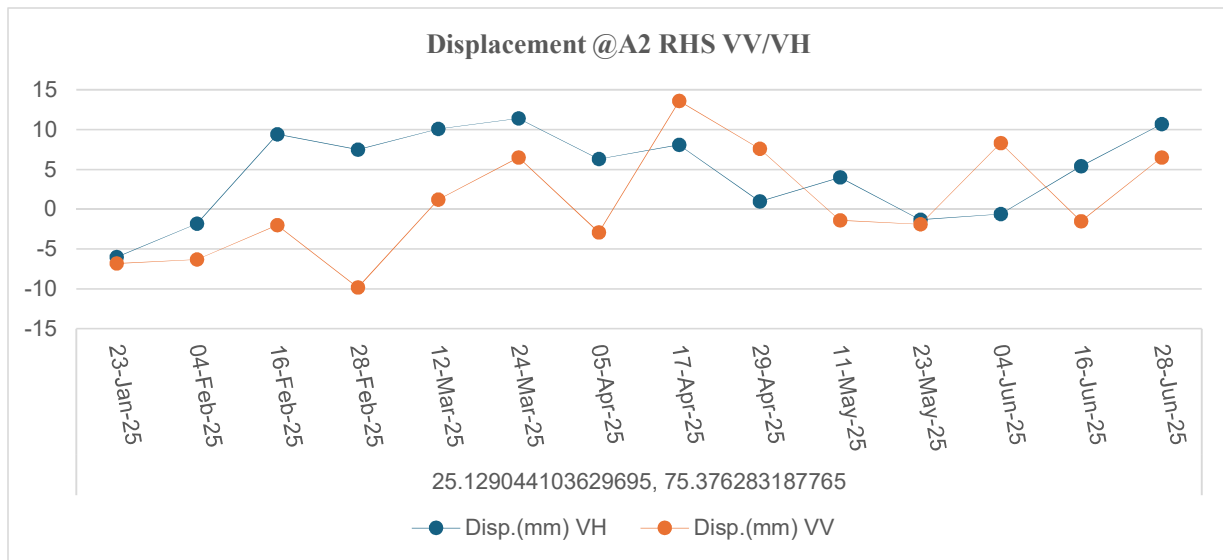
Critical Events & Anomalies:

- **January 2025:** VH begins around +7 mm, while VV is approximately –12 mm, indicating initial lateral displacement accompanied by noticeable vertical settlement.
- **June 2025:** VV declines rapidly from –8 mm to –22 mm, while VH remains elevated between +15 mm and +18 mm, highlighting severe vertical settlement occurring alongside sustained lateral displacement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.129044103629695, 75.376283187765)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** exhibits alternating uplift and settlement behaviour, with values ranging from approximately -10 mm to +14 mm.
- **Horizontal (VH) displacement** at A2 RHS shows moderate variability with an overall positive tendency, ranging approximately from -6 mm to +11 mm.

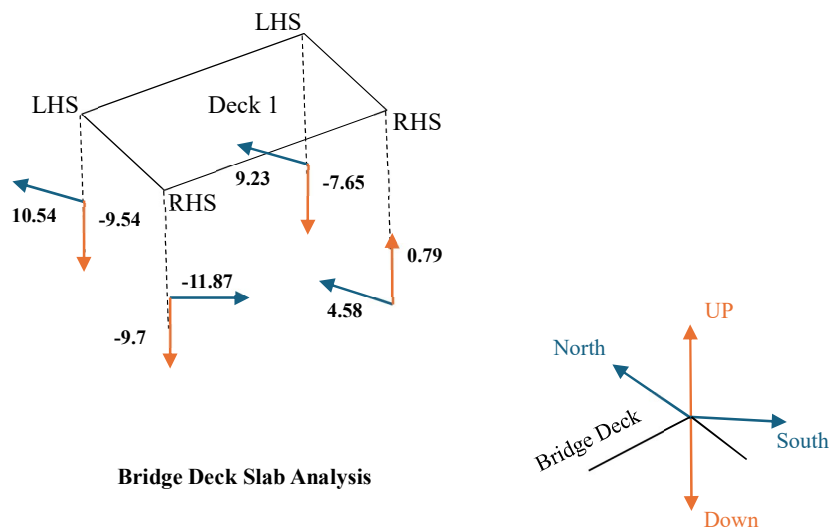
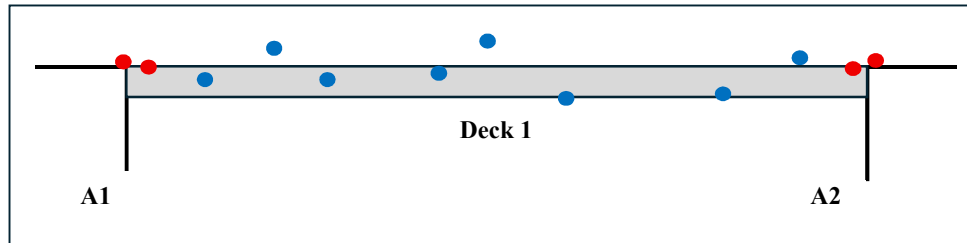
Critical Events & Anomalies:

- **January 2025:** VH starts near -6 mm, while VV is around -7 mm, indicating initial combined lateral and vertical settlement.
- **April 2025:** VV spikes to approximately +14 mm while VH remains positive (+8 mm), marking the maximum vertical uplift event in the dataset.
- **June 2025:** VH rises again, ending near +10 mm, while VV fluctuates between -2 mm and +8 mm, suggesting renewed lateral displacement with unstable vertical response.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits pronounced non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The dominance of settlement over uplift confirms overall downward bending with localized vertical rotation, rather than uniform vertical translation

Horizontal Displacement Behaviour:

- Horizontal displacement shows strong directional variability across the deck width, clearly reflecting torsional deformation behaviour. The uneven distribution of lateral displacement across LHS and RHS confirms torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-influenced deformation response, characterized by dominant vertical settlement combined with significant asymmetric horizontal displacement. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and correlation with loading conditions, bearing performance, and boundary restraints.